

Table 9-2: *Math Operator Symbols, Meanings, and Magic Method Names*

Symbol	Meaning	Magic method name
+	Addition	<code>__add__()</code>
-	Subtraction	<code>__sub__()</code>
*	Multiplication	<code>__mul__()</code>
/	Division (floating-point result)	<code>__truediv__()</code>
//	Integer division	<code>__floordiv__()</code>
%	Modulo	<code>__mod__()</code>
abs	Absolute value	<code>__abs__()</code>

For example, to handle the + operator, you would implement a method in your class like this:

```
def __add__(self, oOther):  
    # Your code here to determine what happens when code  
    # attempts to add two of these objects.
```

A full list of all the magic or dunder methods can be found in the official documentation at <https://docs.python.org/3/reference/datamodel.html>.

Vector Example

In math, a *vector* is an ordered pair of x and y values that is often represented on a graph as a directed line segment. In this section, we'll build a class that uses math operator magic methods to operate on vectors. There are a number of math operations that can be performed on vectors. [Figure 9-2](#) shows an example of adding two vectors.